



Centre for Advanced Studies
(A Research Driven Autonomous Government Institution)
Dr. A.P.J. Abdul Kalam Technical University Uttar Pradesh,
Formerly Uttar Pradesh Technical University
Sec-11, Jankipuram Vistar, Lucknow, Uttar Pradesh, India
PIN Code-226031



Recruitment for Faculty Positions (Assistant Professor on contract)

The Centre for Advanced Studies (An in-campus research driven Government institute of Dr. A. P. J. Abdul Kalam Technical University Uttar Pradesh Lucknow) requires Faculty members (On Contractual Basis with honorarium of Rs. 70000/- per month) for handling academic activities in following programs.

- (1) Computer Science & Engineering**
- (2) Mechatronics**
- (3) Nanotechnology**
- (4) Manufacturing Technology & Automation**
- (5) Energy Science and Technology programs.**

The program wise details of the vacancies are as per following details:

S No.	Program Name	No. of vacancies
1	Computer Science & Engineering	3
2	Mechatronics	1
3	Nanotechnology	2
4	Manufacturing Technology & Automation	1
5	Energy Science & Technology	1
		8 (Eight)

Note: The reservation shall be applicable as per UP Government rules.

The interested candidates meeting the eligibility criteria are required to come for interview on 14-16 July 2022 (as per following schedule) at the following venue. The interested and eligible candidates should send the online applications on career@cas.res.in in prescribed proforma (attached as Annexure-1 in the last of the advertisement available on <http://cas.res.in/>) addressing to The Director, Centre for Advanced Studies along with supportive academic documents and relevant experience certificates. The subject of email should be “Application for the post of Assistant Professor (on contract) in (Name of Department)”.

The interviews are scheduled as per following schedule:

S. No.	Program Name	Date & Time	Venue
1	Computer Science & Engineering	14 th July, 2022 10:30AM	Director Office Centre for Advanced Studies, Dr. APJ Abdul Kalam Technical University, Uttar Pradesh, Sector-11, Jankipuram Extension, Lucknow- 226031
2	Mechatronics	15 th July, 2022 10:30AM	
3	Manufacturing Technology and Automation		
4	Energy Science & Technology	16 th July, 2022 10:30AM	
5	Nanotechnology		

Venue of interview:

Director office,
Centre for Advanced Studies,
Dr. A. P. J. Abdul Kalam Technical University Uttar Pradesh, Sector-11, Jankipuram
Extension.

For updates and corrigendum please visit the following website regularly.

<https://cas.res.in/index.html> and <https://aktu.ac.in/>

Honorarium:

Assistant Professor (Contractual):

- Rs. 70000/- p.m. (Consolidated) for PhD degree holders in relevant field of Science/ Engineering
- Rs. 45000/- pm (consolidated) for the candidates who have submitted the PhD thesis for the award.

Note: No TA/DA shall be paid to the candidates for appearing in the interview. The expenses shall be borne by the candidates themselves.

Minimum Eligibility Criteria

Minimum eligibility is PhD degree in relevant branch of Science/Engineering, however, PhD thesis submitted for the award may also be considered with monthly honorarium of Rs. 45000/-.

The selection of candidates will be based on written examination followed by interview of written qualified candidates.

Detailed Academic qualification (program wise)

A. Assistant Professor (Contractual) – Mechatronics

All candidates having

(1) B.E./B.Tech and M.E./M.Tech in Mechanical engineering/Electronics and Instrumentation Engineering/ Mechatronics/Robotics/Computer Engineering and related branches, with First class or equivalent either in B.E./B.Tech or M.E./M.Tech AND

Ph.D. in relevant branch of Engineering and Technology.

(2) At least two papers should be SCI/ SCOPUS/ CiteSeerX/ DBLP indexed or published in journal of international repute

(3) Education degree(s) from institutes of national and international importance

Note: Candidates who have done Ph.D. after the Bachelor's Degree from institution of National importance with GATE shall be eligible for the post of Assistant Professor.

B. Assistant Professor (Contractual) – Nanotechnology

(1) All candidates having B.E./B.Tech and M.E./M.Tech with 1st class or equivalent either in B.E./B.Tech or M.E./M.Tech in any branch of Engineering and Technology or any related field

OR

B.Sc. and M.Sc. degree in any branch of Science like Physics, Chemistry, Biology, Mathematics, Materials Science, Electronics or any related field relevant to the area of Nanotechnology with 1st Class or equivalent in B.Sc. and M.Sc. both

AND

Ph.D in relevant branch of Science/Engineering in the area/field relevant to Nanotechnology

(2) At least two papers should be SCI/ SCOPUS/ CiteSeerX/ DBLP indexed or published in journal of international repute

(3) Education degree(s) from any national/international institute/university of repute and standing

Note: Candidates who have done Ph.D. after the Bachelor's Degree from institution of National importance with GATE shall be eligible for the post of Assistant Professor.

C. Assistant Professor (Contractual) – Energy Science and Technology

(1) All candidates having B.E./B.Tech and M.E./M.Tech with 1st class or equivalent either in B.E./B.Tech or M.E./M.Tech in any branch of Engineering and Technology or any related field

OR

B.Sc. and M.Sc. degree in any branch of Science like Physics, Chemistry, Biology, Mathematics, Materials Science, Electronics or any related field relevant to the area of Energy Science and Technology with 1st Class or equivalent in B.Sc. and M.Sc. both

AND

Ph.D in relevant branch of Science/Engineering in the area/field relevant to Energy Science and Technology

- (2) At least two papers should be SCI/ SCOPUS/ CiteSeerX/ DBLP indexed or published in journal of international repute
- (3) Education degree(s) from any national/international institute/university of repute and standing

Note: Candidates who have done Ph.D. after the Bachelor's Degree from institution of National importance with GATE shall be eligible for the post of Assistant Professor.

D. Assistant Professor (Contractual) – Manufacturing Technology and Automation

- (1) All candidates having
B.E. / B.Tech and M.E/M.Tech in Mechanical Engineering/ /Robotics and Automation/Production Engineering /Manufacturing Technology & Automation/Manufacturing technology and related branches, with First class or equivalent either in B.E./B.Tech or M.E./M.Tech
AND
Ph.D. in relevant branch of Engineering and Technology.
- (2) At least two papers should be SCI/ SCOPUS/ CiteSeerX/ DBLP indexed or published in journal of international repute
- (3) Education degree(s) from institutes of national and international importance

Note: Candidates who have done Ph.D. after the Bachelor's Degree from institution of National importance with GATE shall be eligible for the post of Assistant Professor

E. Assistant Professor (Contractual) – Computer Science & Engineering

- (1) All candidates having
B.E./B.Tech and M.E./M.Tech with 1st class or equivalent either in B.E./B.Tech or M.E./M.Tech in Computer Science and Engineering/ Information Technology/ Electronics Engineering/ Electronics and Communication Engineering branch
AND
Ph.D in Computer Science and Engineering/ Information Technology/ Electronics Engineering/ Electronics and Communication Engineering branch
- (2) At least two papers should be SCI/ SCOPUS/ CiteSeerX/ DBLP indexed or published in journal of international repute
- (3) Education degree(s) from any national/international institute/university of repute

Note: Candidates who have done Ph.D. after the Bachelor's Degree from institution of National importance with GATE shall be eligible for the post of Assistant Professor.

The Candidates are requested to bring the following at the time of interview:

1. A Demand Draft of Rs. 1000 (for Gen/OBC) or Rs. 500 (for SC/ST, PH & woman) in favor of “Finance Officer, AKTU Lucknow” payable at Lucknow as processing fee.
2. Filled in Performa (attached in last) along with the supportive documents.
3. Two recent photographs (passport size).
2. All academic and experience certificates in original (one set of self-attested photocopies of the same).
3. One government issued ID proof in original and photocopy.

NOTE: The candidates who are unable to come on the dates of interview at the venues can send their resumes on career@cas.res.in so that if required they can be called during second round of recruitment process.

Syllabus for written exam

There shall be a written objective examination of the relevant discipline/ branch.

1. Computer Science & Engineering:

SYLLABUS will be of CS/IT GATE 2022

2. Energy Science & Technology:

Energy Sources, Fuel and Combustion, Energy Conversion and Storage System, Introduction to the Energy materials and Fundamentals of Energy Devices, Energy Audit And Management, Bio Energy Engineering, Hydrogen Energy, Electric Vehicles Technology, Wind Energy Conversion Systems, Nuclear Reactor Theory, Overview of Material Characterization Techniques, Environmental Engineering And Pollution Control, Process Modelling and Simulation in Energy Systems, Thermal Engineering, Energy Audit and Management, Solar Energy Engineering, Waste Management for Energy Generation Technologies, Environmental Impact Assessment and Economic Analysis of Energy Technologies, Instrumentation, Nuclear, Batteries & Fuel Cells, Electrochemistry and Super capacitor, Wind Energy Engineering, Thin Film Technology and Applications, Nuclear Materials, Instrumentation in Energy Systems, Computational Heat Transfer, Power Plant and Electrical Energy Technology, Renewable Power Generation Sources, Energy Systems Modelling and Analysis.

3. Manufacturing Technology & Automation:

Material Science including composite materials, Strength of Material, Manufacturing Technology, Metal removal Processes, Theory of Metal Forming, Conventional and unconventional manufacturing processes, Additive Manufacturing, Advanced Foundry

and welding technology, Machine tool control and condition monitoring, Surface Engineering, MEMS, NEMS, Production of Gears and Screws, Friction, wear and lubrication, Precision Engineering, Fluid Mechanics and machinery, Theory of machines and design of machine elements, Materials handling and management, Industrial management, Production planning and control, Time and motion study, Total Quality management, Pneumatics and control, Computer Assisted Manufacturing, Computer Aided Design, Computer Integrated Manufacturing, Flexible Manufacturing System, Finite Element Method, Automobile Engineering, Operation Research, Design Planning and Control of Production Systems, Supply Chain Management, Project Management, Inspection and Testing in Manufacturing, Automation and Robotics, Automation and control, Tool Engineering, Refrigeration and Air conditioning, Simulation and modelling.

4. Nanotechnology:

Introduction to material science, Crystal structure and Types of bonding, Energy bands in metals, insulators and semiconductors, electrical and thermal properties of materials, metals, ceramics, polymers, composites and nanomaterials, basics of Electrochemistry, diamagnetism, ferromagnetism, para-magnetism, ferrimagnetism and anti-ferrimagnetism, Op-amp, NAND, NOR and XOR gate, simple combinational circuits, Diode and BJT its principles and applications, principles of communication, amplitude and frequency modulation, matrix Algebra, vector algebra and calculus and linear differential equations, strength of materials, heat transfer and fluid mechanics, principles of quantum mechanics, Heisenberg's uncertainty principle and wave particle duality.

5. Mechatronics:

Engineering Materials: Structure and properties of engineering materials, phase diagrams, heat treatment, stress-strain diagrams for engineering materials.

Machining and Machine Tool Operations: Mechanics of machining; basic machine tools.

Metrology and Inspection: Limits, fits and tolerances;

Heat-Transfer: Modes of heat transfer; Machine Design: Design for static and dynamic loading;

Mechatronics Systems, Actuators, Drives, PLCs.

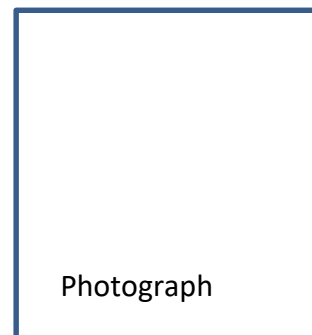
Number representations: binary, integer and floating-point- numbers. Data converters: sample and hold circuits, ADCs and DACs. Basic control system components; Feedback principle; Transfer function; Block diagram representation; Signal flow graph; Transient and steady-state analysis of LTI systems.

Resistive-, capacitive-, inductive-, piezoelectric-, Hall effect sensors and associated signal conditioning circuits; transducers for industrial instrumentation

Note: The questions for faculty aptitude test should be of GATE examination/PG level.

Brief Information Performa

Centre for Advanced Studies
Dr. A.P.J. Abdul Kalam Technical University Uttar Pradesh,
 Formerly Uttar Pradesh Technical University
 Sec-11, Jankipuram Vistar, Lucknow, Uttar Pradesh, India
 PIN Code-226031

**Name:****Father/Husband Name:****Address:****Mobile No.****Email id (in capitals):****Category:** Gen OBC SC ST**Gender:** Male Female Others**DOB:****Adhaar Number:****Department Name:****Date of Interview:****Academic Qualification:**

S No.	Qualification	Branch/ Subjects	Name of Institute	Board/ Univ.	Passing Year	Percentage/C GPA	Division
1	PhD						
2	M.Tech/M.Sc.						
3	B.E./B.Tech/B.Sc.						
4	12						
5	10						

Research Paper (SCI/SCIE only)

S No.	Authors	Are you corresponding/First author	Journal name	Publisher	Vol. No. & Page No.	Impact Factor	Year of publication

Research Papers (Scopus indexed)

S No.	Authors	Are you corresponding/First author	Journal name	Publisher	Vol. No. & Page No.	Impact Factor	Year of publication

Conference proceedings (Indexed in SCI/SCIE/Scopus only)

S No.	Authors	Are you corresponding /First author	Conference title & place	Organizing Institute	Year of publication

Research Papers/conferences (Non SCI/non-SCIE/Non Scopus indexed)

S No.	Authors	Are you corresponding/First author	Journal name	Publisher	Vol. No. & Page No.	Impact Factor	Year of publication

Book/ Book Chapter (International publisher)

S No.	Title of Book/ Book chapter	Publisher	ISBN	Remark

Book/ Book Chapter (Indian publisher)

S No.	Title of Book/ Book chapter	Publisher	ISBN	Remark

Consultancy and Funded Research projects

S No.	Title of project	Funding agency	Cost (Rs)	Year	Status (Completed/ ongoing)

Patents

S No.	Title	Category (US/ Indian)	Year	Status (Awarded/ applied)	Amount of Revenue generation (Rs)

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Work experience (In reverse chronological order)

S No.	Post hold	Institute name	Duration	Salary drawn	Reason of leaving

Ph.D. / M.Tech supervision by candidate

S No.	PhD/ M.Tech	Title	As guide/ Guide	Main Co-	Institute Name	Year of award

Experience of conducting Conference/ Workshop/ Seminar/ FDPs/ Short term courses as convener/ co-convener

S No.	Title of event	Institute	Convener/ Co-convener	Duration	No. of participants	Remark

Declaration: I hereby declare that the information furnished by me are to the best of my knowledge and if found false at any time or during my service my candidature will be cancelled.

Place:

Name & Sign of candidate

Date: